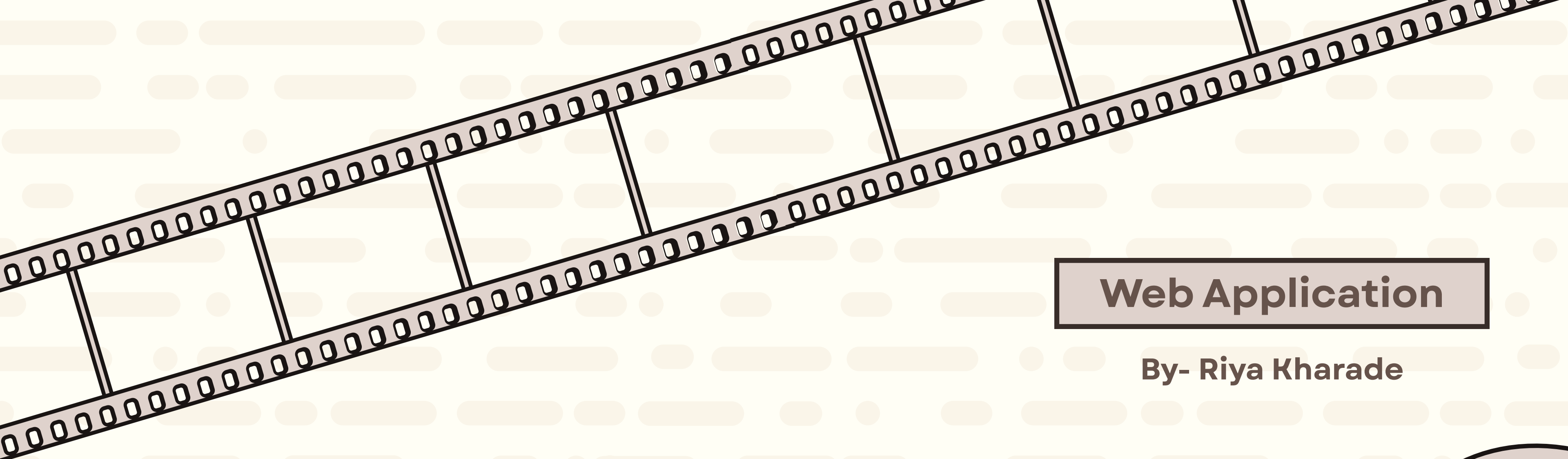


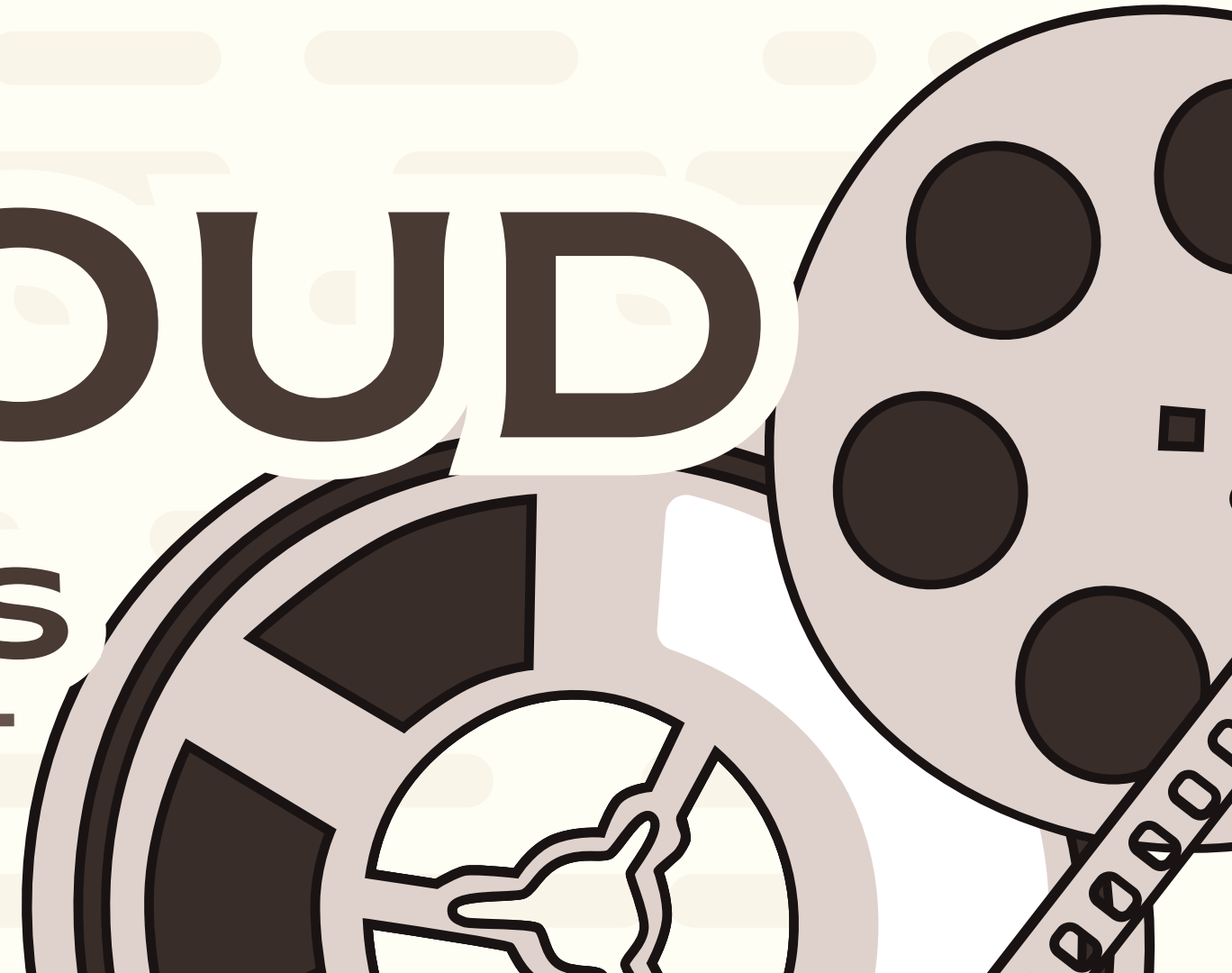


Web Application

By- Riya Kharade

CHITRA CLOUD

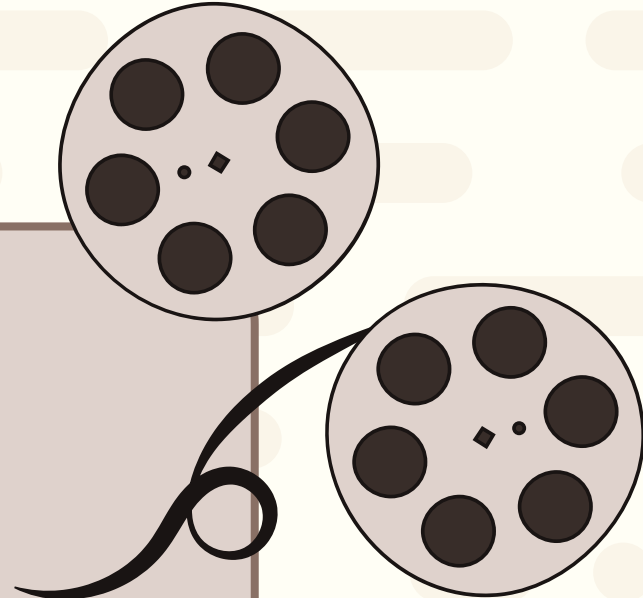
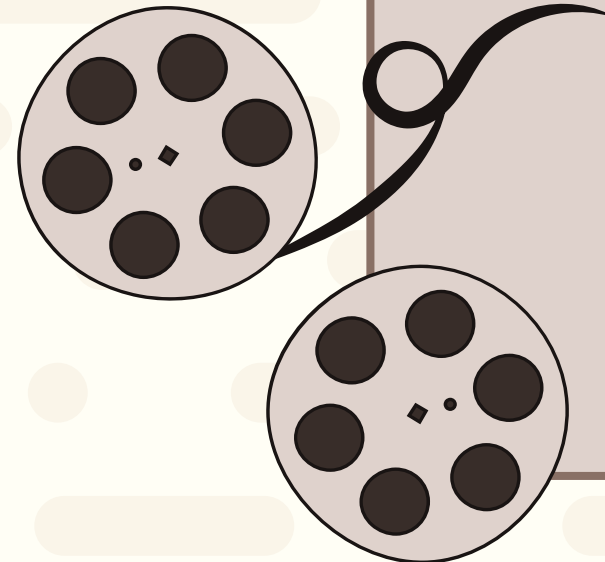
Using AWS Services





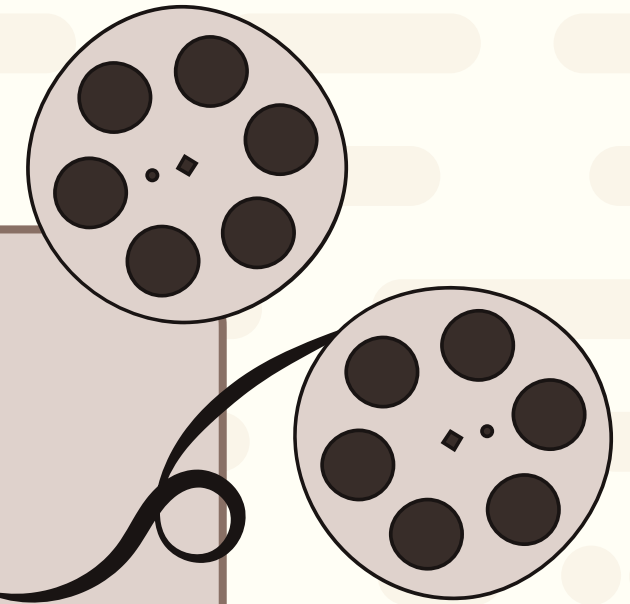
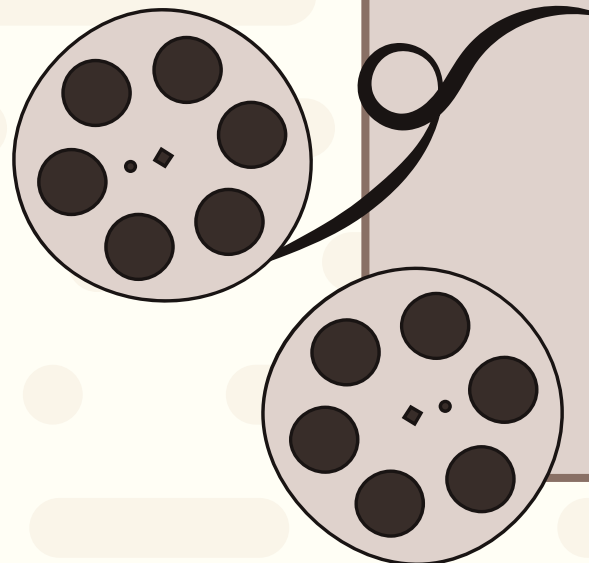
Problem Statement

Existing image editing tools are complex, mostly paid, and require multiple platforms for tasks like PDF to image conversion. This creates a need for a simple, free, and all-in-one solution with an easy-to-use interface.



Introduction

ChitraCloud is a cloud-based image editing web application that allows users to upload, edit, convert, and store images with a simple and user-friendly interface. It also includes PDF to image conversion and uses AWS services for secure and efficient operation.



Features Of Project

Image Filters

Apply different filters to enhance image quality.

Adjustments

Control brightness, contrast, and more.

Resizing

Change image size and aspect ratio.

Transform

Rotate and flip images easily.

Gallery

View, edit, delete, and manage images.

Conversion

Convert PDF files into images.

AWS Services Used

- Amazon S3
- Amazon DynamoDB
- Amazon Cognito
- AWS IAM





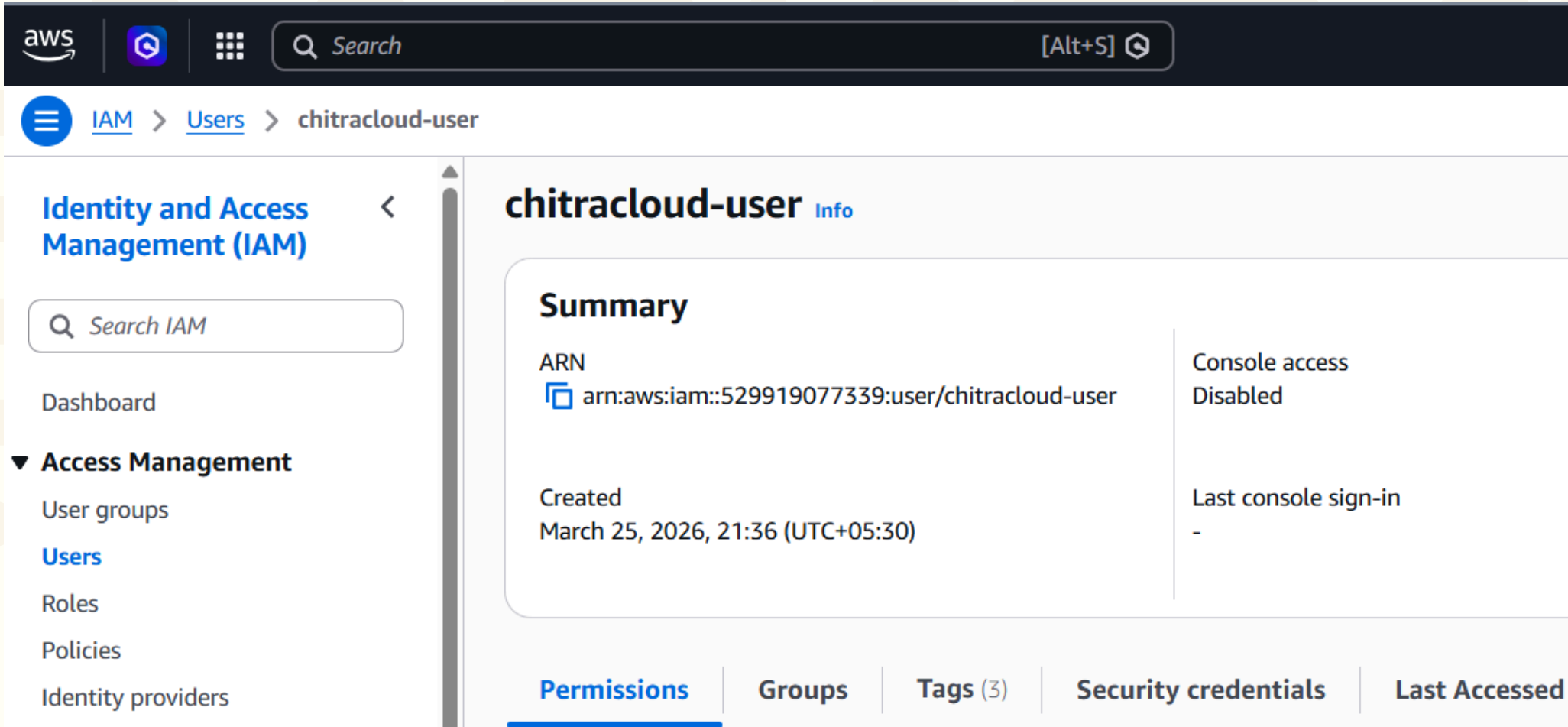
Amazon IAM

What it does:

Manages access and permissions for AWS services.

Work in Project:

- Controls access to S3 and DynamoDB
- Provides secure permission to services
- Ensures only authorized actions are allowed



Identity and Access Management





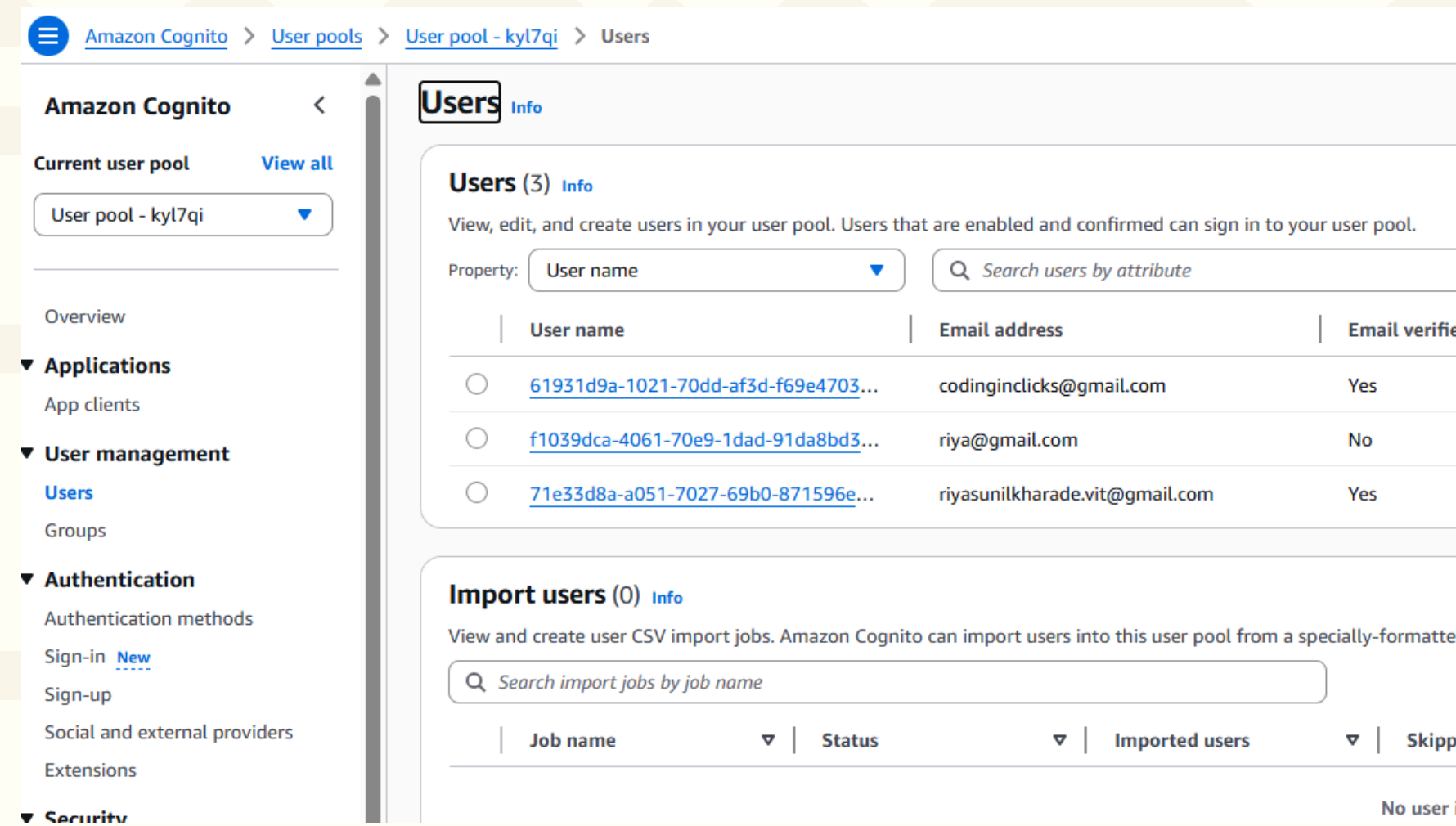
Amazon Cognito

What it does:

Handles user authentication and login system.

Work in Project:

- User signup and login
- OTP verification
- Secure user authentication
- Manages user sessions



Amazon Cognito





Amazon S3

What it does:
Provides cloud storage for files and images.

Work in Project:

- Stores uploaded images
- Stores edited images
- Allows image retrieval anytime
- Ensures secure and scalable storage

chitracloud-riya-210

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (4)

Copy S3 URICopy URLDownloadOpen ↗DeleteActions▼Create folc

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions.

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	 1775639895375-134182398316117465.jpg	jpg	April 8, 2026, 14:48:17 (UTC+05:30)	2.0 MB	Standard
☐	 1775640051484-134191054033231267.jpg	jpg	April 8, 2026, 14:50:53 (UTC+05:30)	2.6 MB	Standard
☐	 1775719468110-134172467560936381.jpg	jpg	April 9, 2026, 12:54:29 (UTC+05:30)	1.6 MB	Standard
☐	 1775719531057-edited.png	png	April 9, 2026, 12:55:32 (UTC+05:30)	11.7 MB	Standard

Simple Storage Service





Amazon DynamoDB

What it does:

Provides a NoSQL database for storing data.

Work in Project:

- Stores image metadata
- Stores upload time
- Supports rename functionality
- Supports delete operations
- Contact Page Details Stored

Tables (2)

Filter by tag

Any tag key

Filter by tag value

Any tag value

Find tables

< 1 >

⚙

☐ ChitraCloudContacts

☒ ChitraCloudImages

ChitraCloudImages

Autopreview

View table details

▼ Scan or query items

☒ Scan

☐ Query

Select a table or index

Table - ChitraCloudImages

Select attribute projection

All attributes

► Filters - optional

Run

Reset

✔ Completed · Items returned: 2 · Items scanned: 2 · Efficiency: 100% · RCUs consumed: 2

Table: ChitraCloudImages - Items returned (2)

Scan started on April 11, 2026, 00:09:39

⌂

Actions

Create item

< 1 >

⚙

☐	id (String)	name	size	type	uploadedAt	url	userId
☐	ad409017-f075-4cd6...	edited.png	12279892	edited	2026-04-09...	https://chit...	riyasunilkharade.vit@gmail.com
☐	f5072f0e-3bd0-48a6...	134172467...	1708093	original	2026-04-09...	https://chit...	riyasunilkharade.vit@gmail.com

Amazon DynamoDB





Why S3 & DynamoDB Together?

Different Roles

- Amazon S3 → Stores image files
- Amazon DynamoDB → Stores image details (metadata)

Limitations of Using One

- S3 cannot efficiently manage or search data
- DynamoDB cannot store large files like images

Final Solution

- Combine both for efficient storage and data management
- Provides faster access and better performance





Future Scope & Conclusion

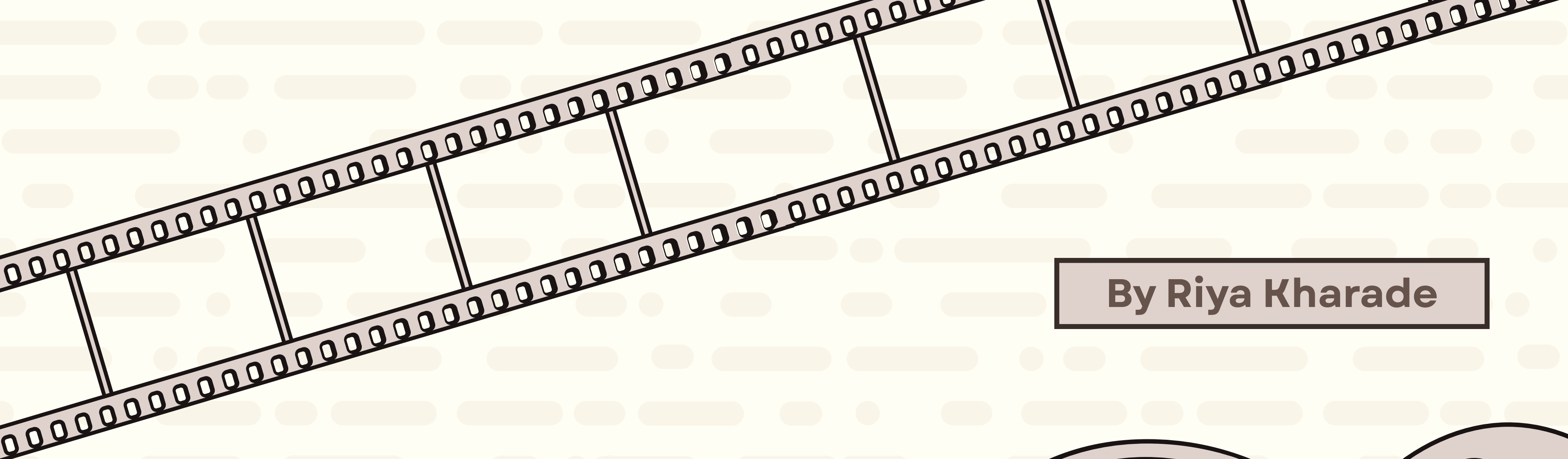
Future Scope

Can be enhanced by adding AI-based editing, mobile support, and advanced features for better user experience.

Conclusion

Successfully developed a cloud-based image editing application using AWS with secure, scalable, and user-friendly features.





By Riya Kharade

Thank You

